

TOPIC 9: POLYMERS

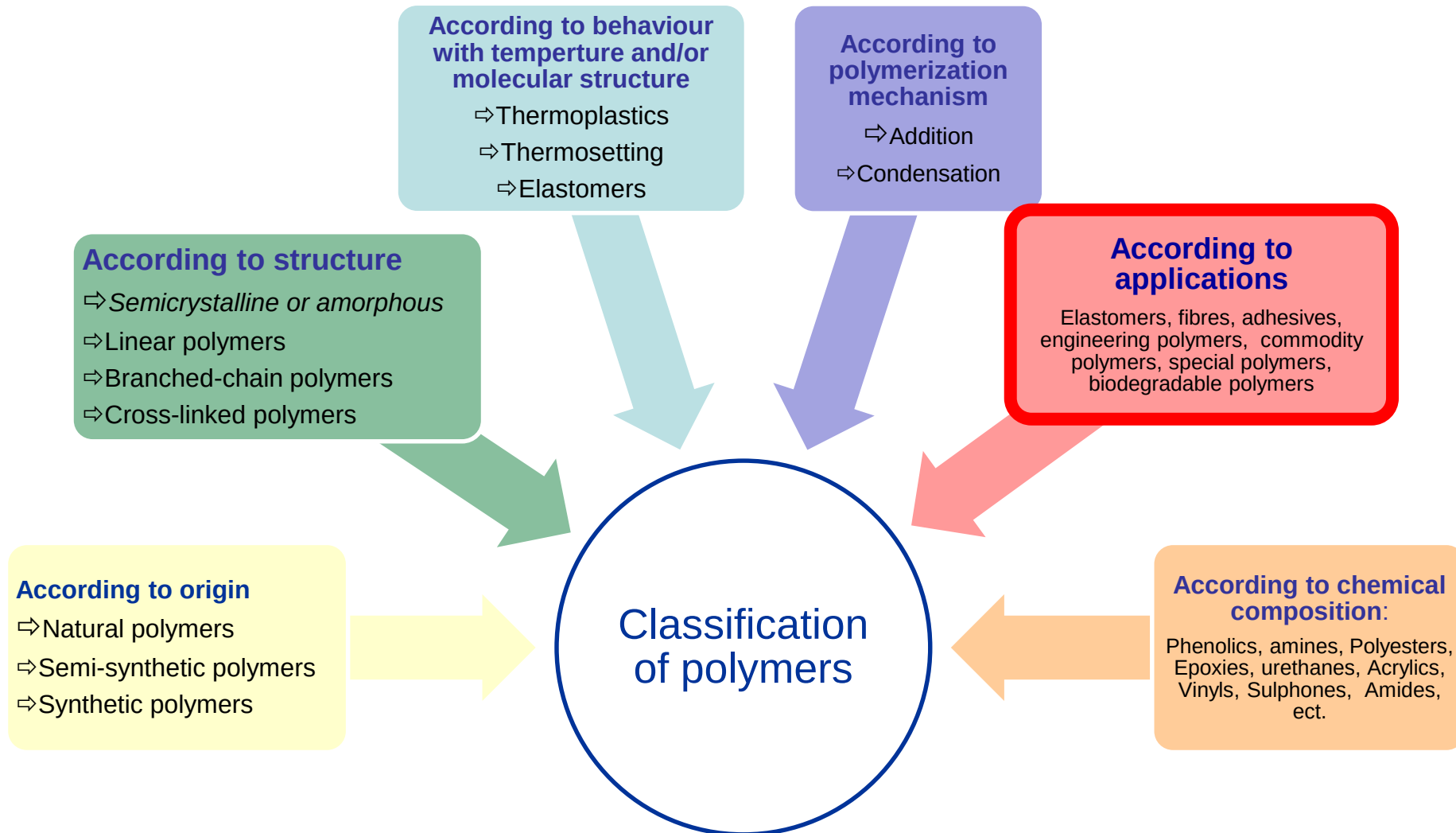
Polymeric Materials II

1. Polymer classification

- 1.1. Thermoplastics
- 1.2. Thermosets
- 1.3. Elastomers

2. Polymer processing

CLASIFICATION

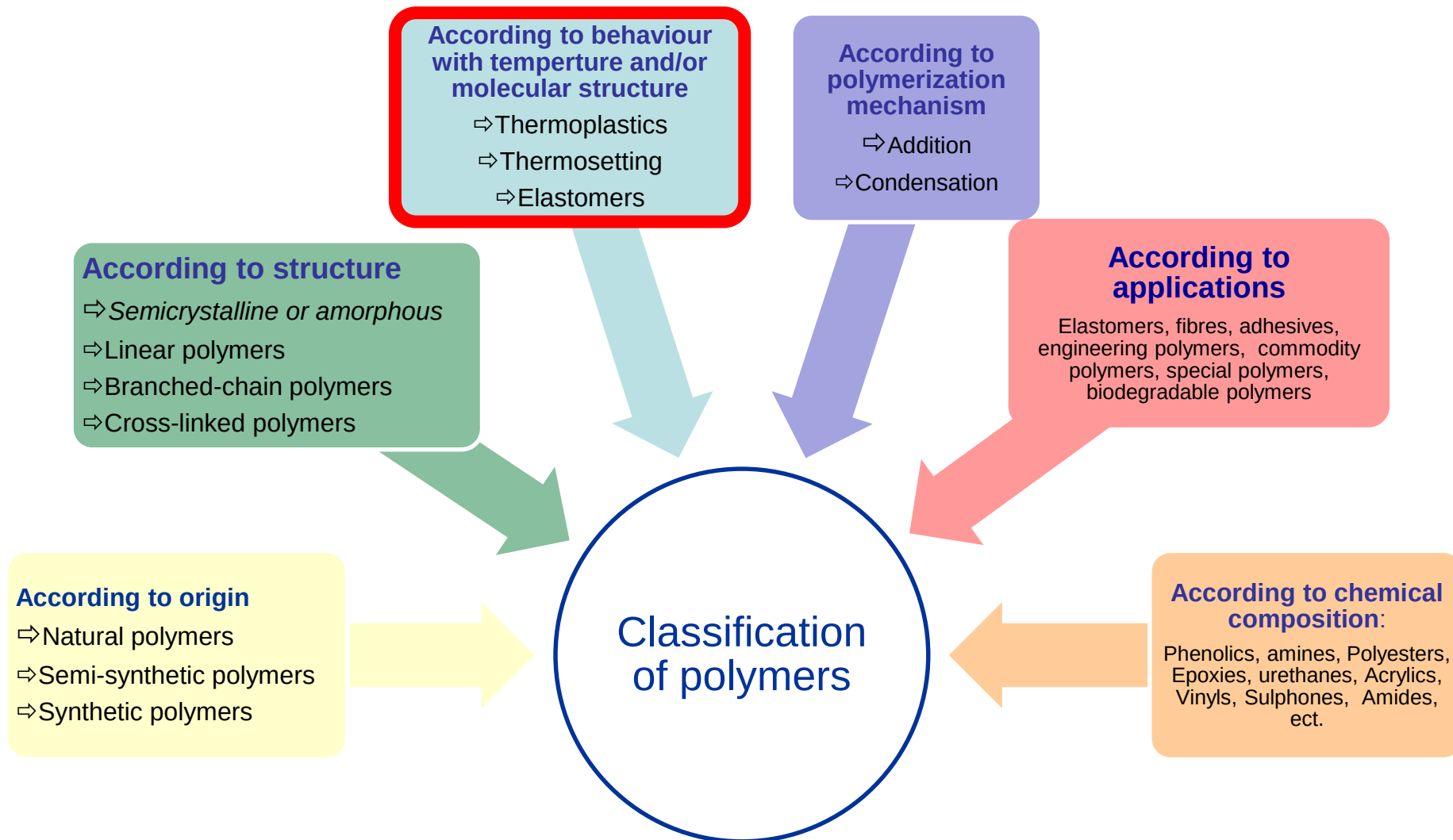


CLASSIFICATION OF POLYMERS

According to applications

- ❑ **Polymers of general use (Commodity):** Polymers of high consumption.
- ❑ **Engineering Polymers:** Good properties between 0-100 °C.
- ❑ **Special Polymers:** High price polymers with outstanding characteristics
- ❑ **Fibers:** generally thermoplastic polymers whose macromolecular chains are easily oriented to the direction of viscous flow during the processing.
- ❑ **Elastomers:** These are polymers (thermoplastics or lightly cross-linked thermosets) that have an elastic deformation > 200%.
- ❑ **Adhesives:** Are substances that combine a high adhesion and a high cohesion

CLASIFICATION



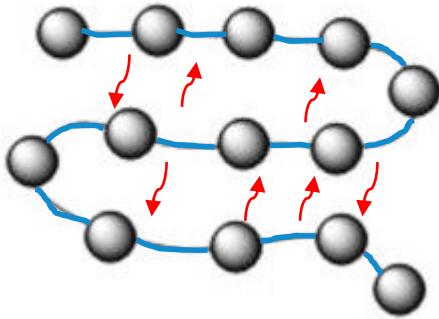
CLASSIFICATION OF POLYMERS

According to their structure and behaviour with temperature

- **Thermoplastics:** They flow when heated and harden again when cooled. Linear or branched polymers where chains of molecules are not interconnected to one another (few or none crosslinks).
- **Thermosetting polymers:** They chemically decompose (break down) when heated, instead of flowing. They have a heavily cross-linked to produce a strong three dimensional network structure.
- **Elastomers:** They have very low modulus of elasticity and high extensibility. These are polymers (thermoplastics or lightly cross-linked thermosets) that have an elastic deformation $> 200\%$.

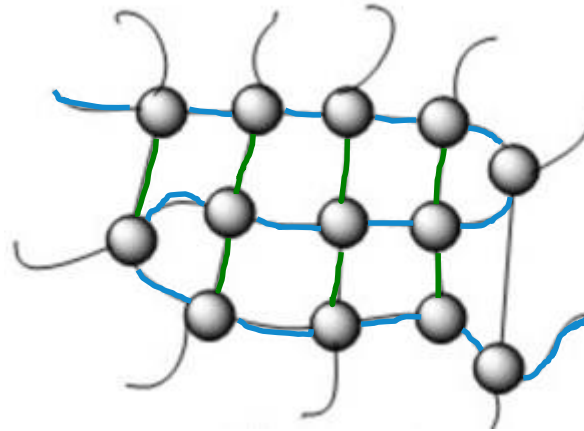
CLASSIFICATION OF POLYMERS

Thermoplastics



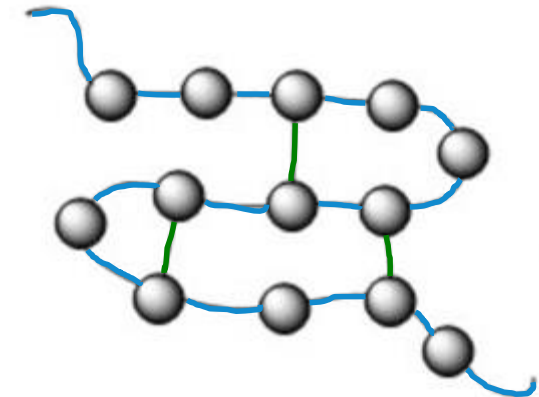
Van der Waal forces

Thermosetting



↑↑↑ Crosslinking

Elastomers



~ Crosslinking

Bonding Strength: **Crosslinking** >> Van der Waal forces

Amorphous or
Semicrystalline

Amorphous

Amorphous

Melt

Do not melt

Do not melt

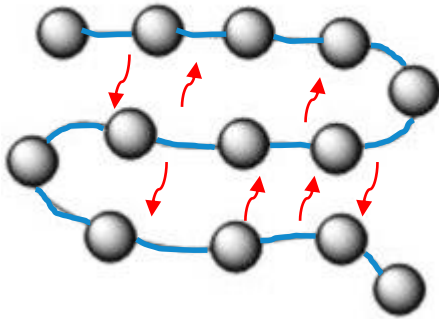
Soluble

Insoluble

Insoluble

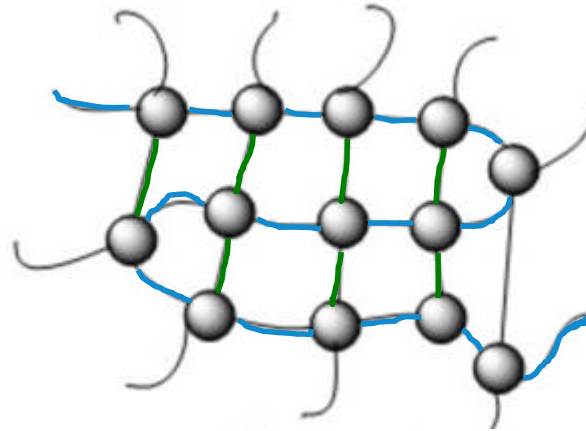
CLASSIFICATION OF POLYMERS

Thermoplastics



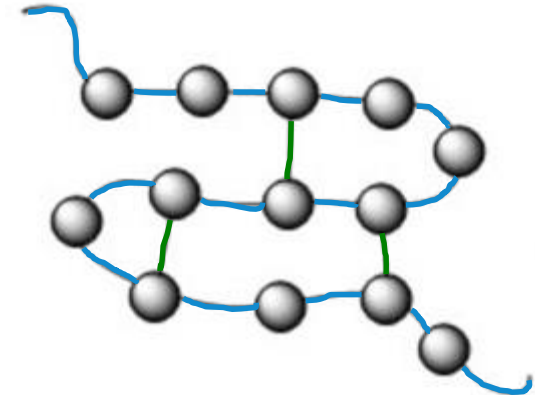
Flow when heated and
harden again when cooled

Thermosetting



chemically
decompose when
heated

Elastomers



CLASSIFICATION OF POLYMERS

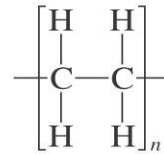
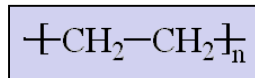
Thermoplastics

“Commodity”

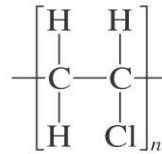
Commonly used thermoplastics



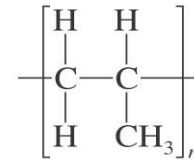
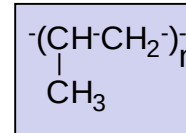
Recycling symbol for thermoplastic commodities



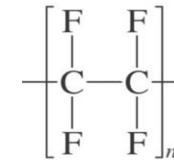
Polyethylene
mp: 110–137°C
(230–278°F)



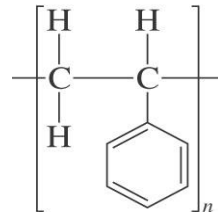
Polyvinyl chloride
mp: ~204°C
(~400°F)



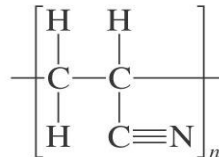
Polypropylene
mp: 165–177°C
(330–350°F)



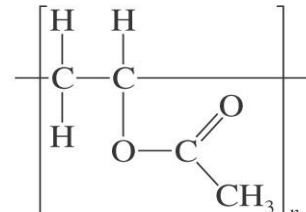
Polytetrafluoroethylene
Softens at 370°C (700°F)



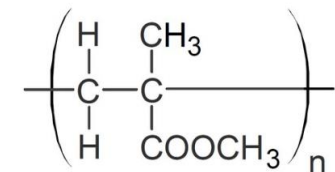
Polystyrene
mp: 150–243°C
(330–470°F)



Polyacrylonitrile
(does not melt)



Polyvinyl acetate
mp: 177°C (350°F)

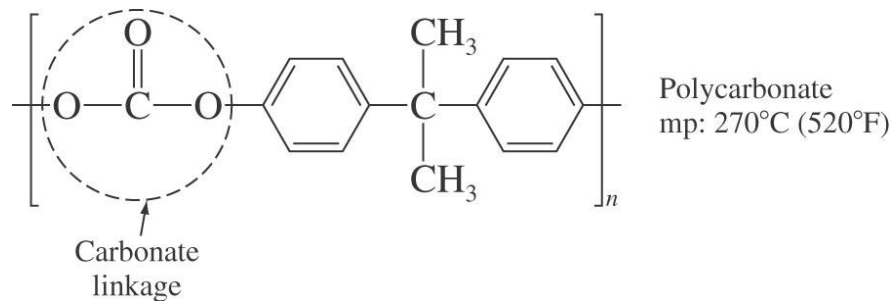
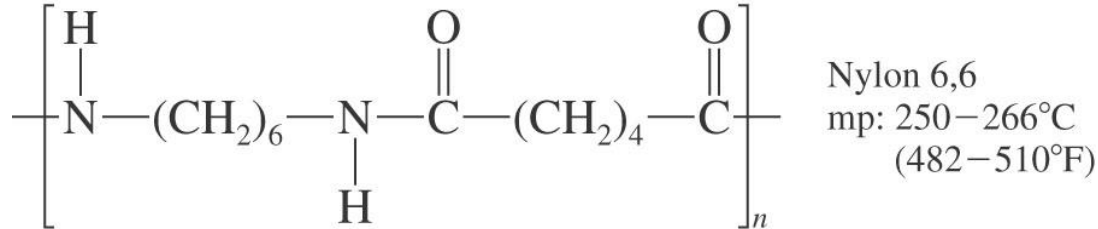
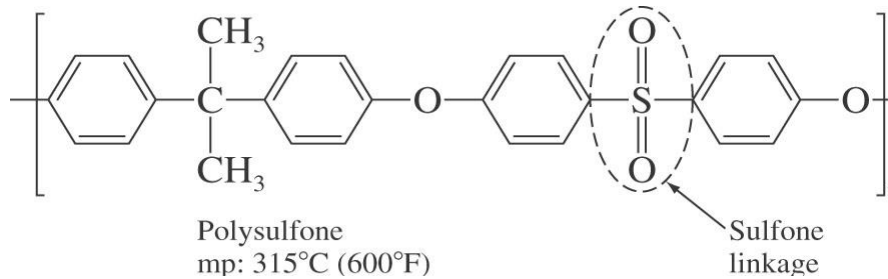


Polymethyl methacrylate: PMMA

CLASSIFICATION OF POLYMERS

Thermoplastics

“Engineering thermoplastics”



CLASSIFICATION OF POLYMERS

Thermoplastics

“Commodity”



POLYMERS

1 – PET

2 – HDPE

3 – PVC

4 – LDPE

5 – PP

6 - PS

CHARACTERISTICS

Transparent, flexible

Rigid, opaque or translucent

Rigid or flexible, shiny

Opaque or transp., very flexible

Translucent, smooth

Transp. or opaque, brittle, shiny

TYPICAL USES

Bottles for carbonated beverage

Bleach bottle

Pipes, bath curtains

Plastic bags

Caps and plugs, syringe

Food packaging

CLASSIFICATION OF POLYMERS

Thermosets

- Crosslinked structure (via covalent bonds)
- Do not melt and are insoluble $\Rightarrow$ cannot be re-cycled
- Formability: heat in mold (thermoheating) and produce chemical reaction $\Rightarrow$ crosslinking

Some advantages in engineering applications

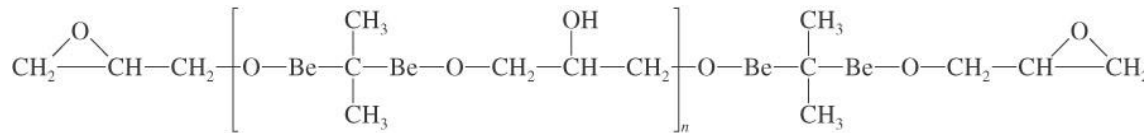
- High thermal stability
- High stiffness
- High dimensional stability
- Resistance to thermal creep and deformation under low loads
- Light weight
- High electrical and thermal insulation.

Epoxy resins (EP) , Polyurethanes (PU), Phenolic resins ...

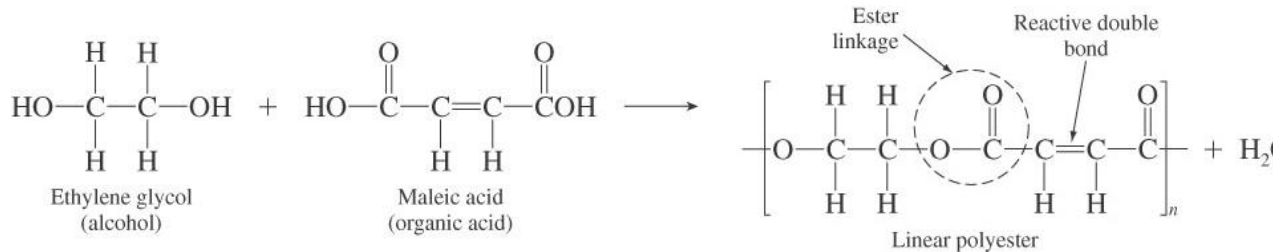
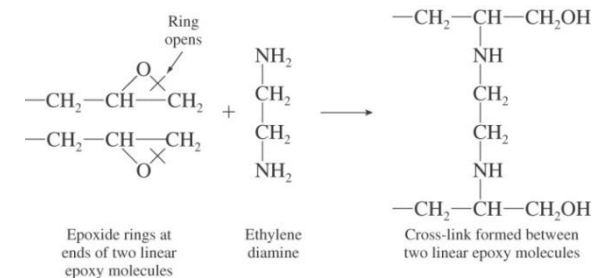
CLASSIFICATION OF POLYMERS

Thermosets

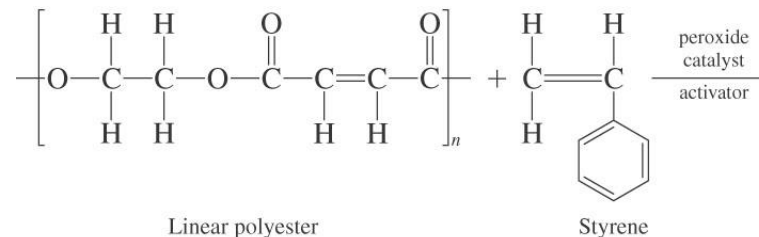
Examples



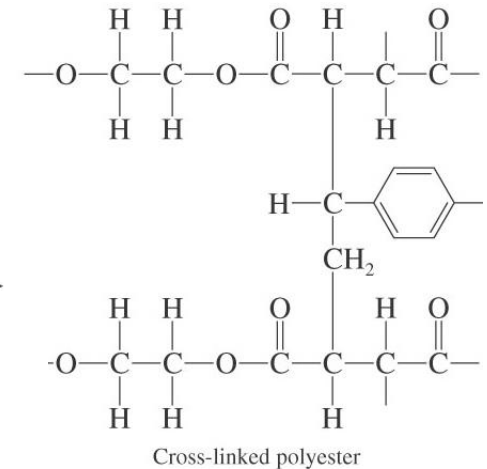
Epoxy resin, Be: benzenic ring



Linear polyester

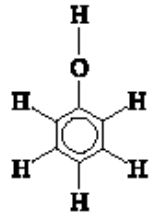
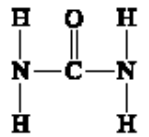
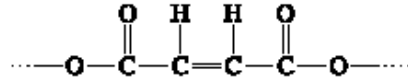
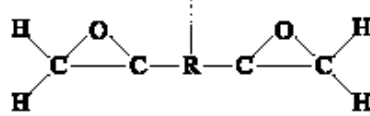
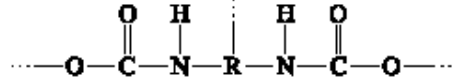
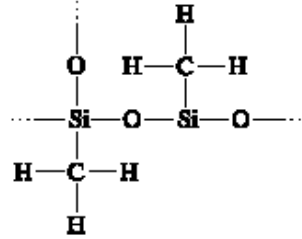


Crosslinked polyester



CLASSIFICATION OF POLYMERS

Thermosets

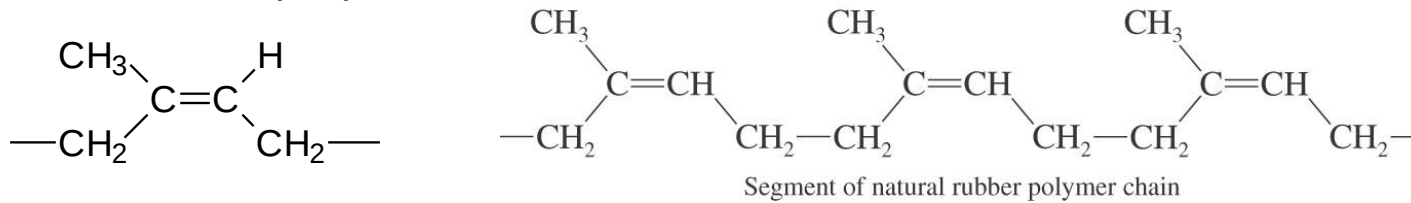
Polymer	Functional Units	Typical Applications
Phenolics		Adhesives, coatings, laminates
Amines		Adhesives, cookware, electrical moldings
Polyesters		Electrical moldings, decorative laminates, polymer matrix in fiberglass
Epoxies		Adhesives, electrical moldings, matrix for composites
Urethanes		Fibers, coatings, foams, insulation
Silicone		Adhesives, gaskets, sealants

CLASSIFICATION OF POLYMERS

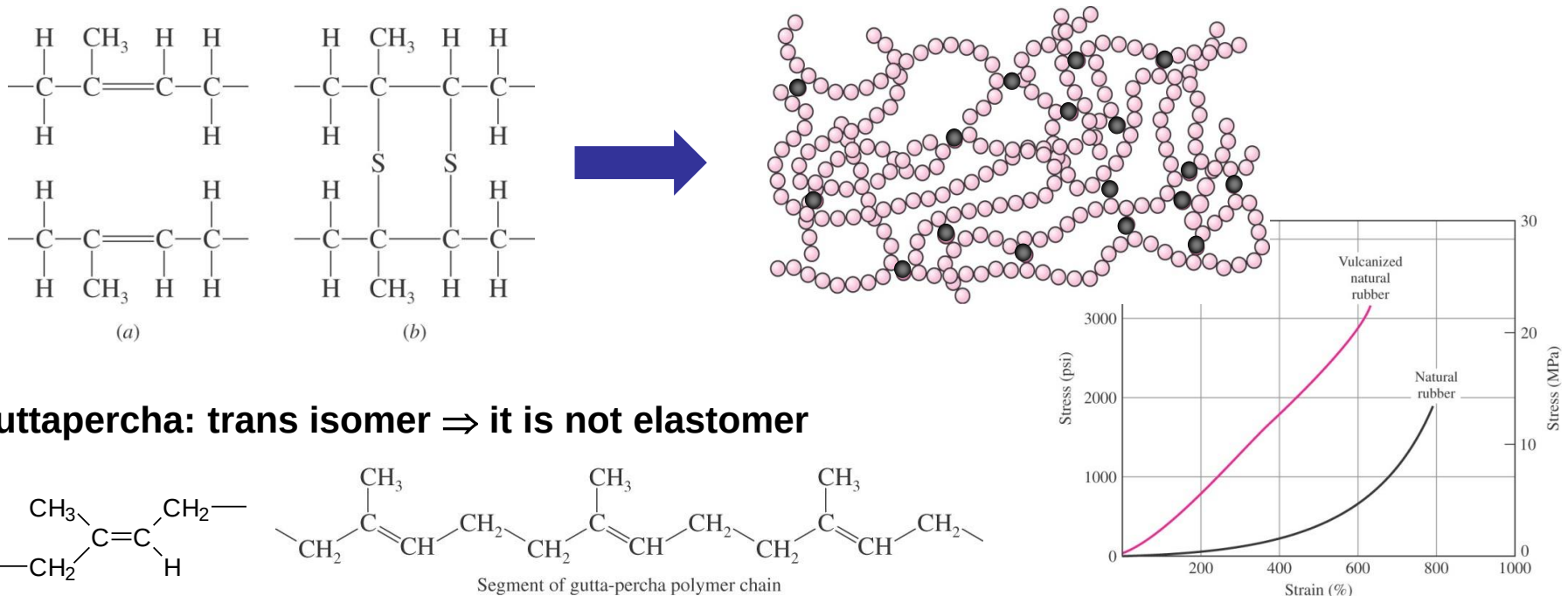
Elastomers

Structures with little crosslinking, insoluble, do not melt, but swell.

Ex. Natural rubber (NR)



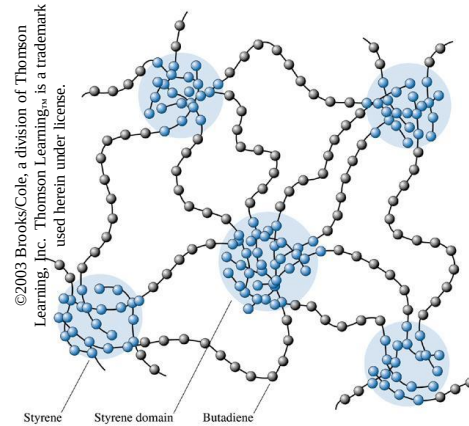
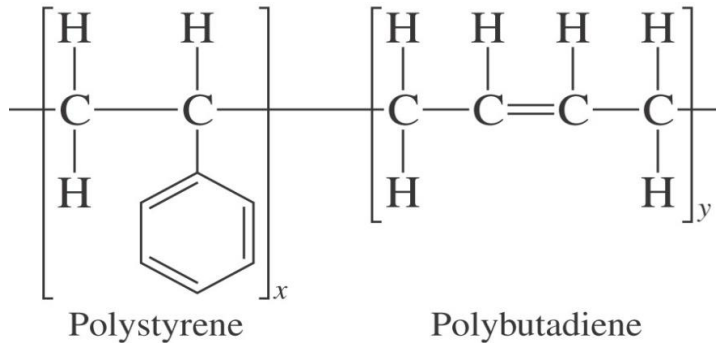
Vulcanization of rubber: crosslinking process with S and Pb carbonate



CLASSIFICATION OF POLYMERS

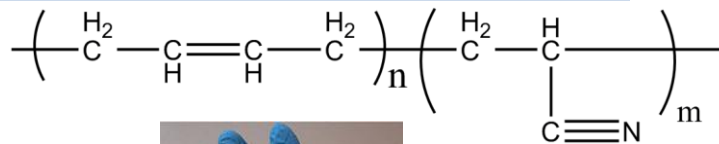
Elastomers

Styrene-Butadiene Rubber (SBR)



styrene-butadiene (SB) copolymer

Acrylonitrile-Butadiene Rubber (NBR)

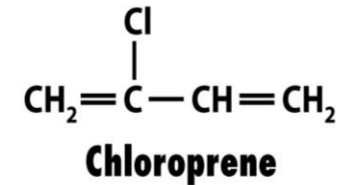


POLYBUTADIENE



Manufacturing of tires consumes about 70% of the world production of polybutadiene.

POLYCHLOROPRENE (NEOPRENE)

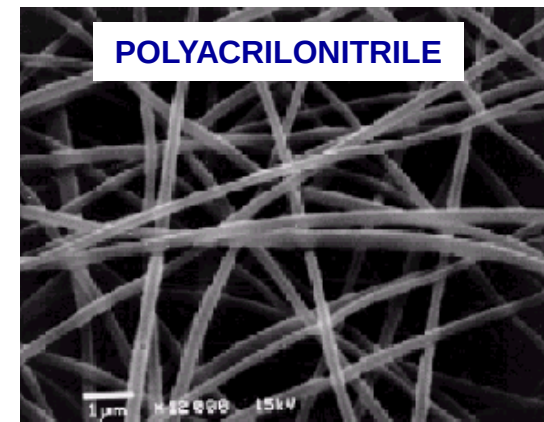
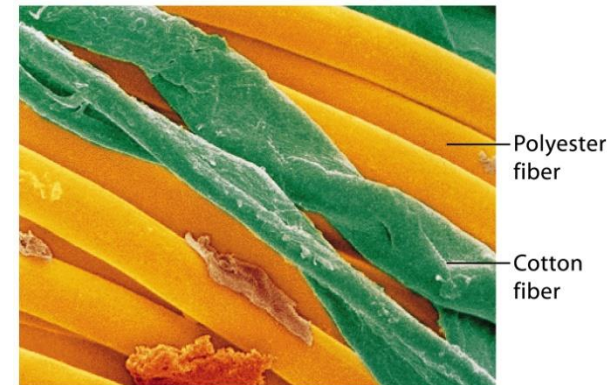


Swinwear fabric made of polychloroprene (neoprene).

CLASSIFICATION OF POLYMERS

Fibres

- Polymers that can be processed in the form of threads with a length to diameter ratio of at least 100:1.
- They are mainly used in the textile industry or as reinforcement in other materials.
- Relatively large MW. L inear chains and crystallinity.
- Physical and chemical properties: high tension strength (in a wide ΔT) and high degree of crystallinity. High E and wear strength and high chemical stability.



POLYMER PROCESSING

- Extrusion
- Injection molding
- Blow molding
- Rotational molding
- Thermoforming
- Compression molding
- Transfer molding
- Foam molding
- Casting

Typical forming processes for thermoplastics:

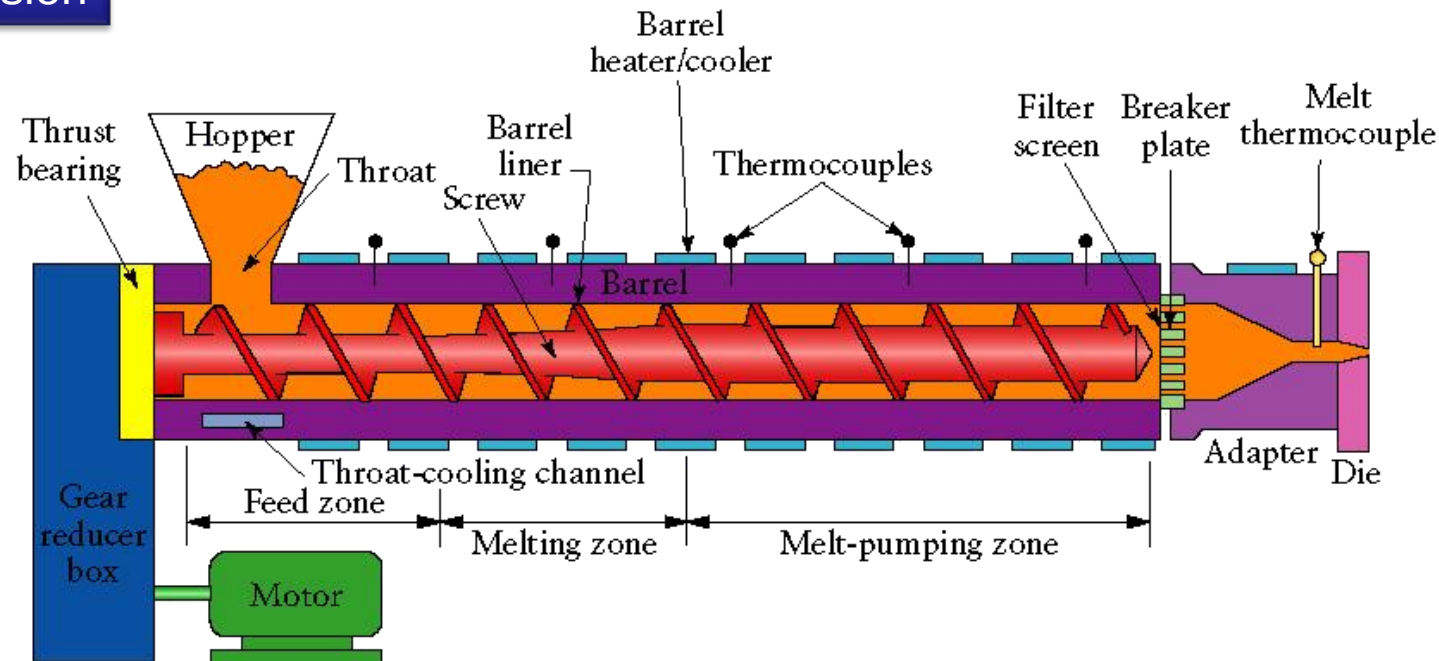
Extrusion ; Blow Molding ; Injection Molding; Thermoforming

Typical forming processes for thermosets:

Compression Molding ; Transfer Molding

POLYMER PROCESSING

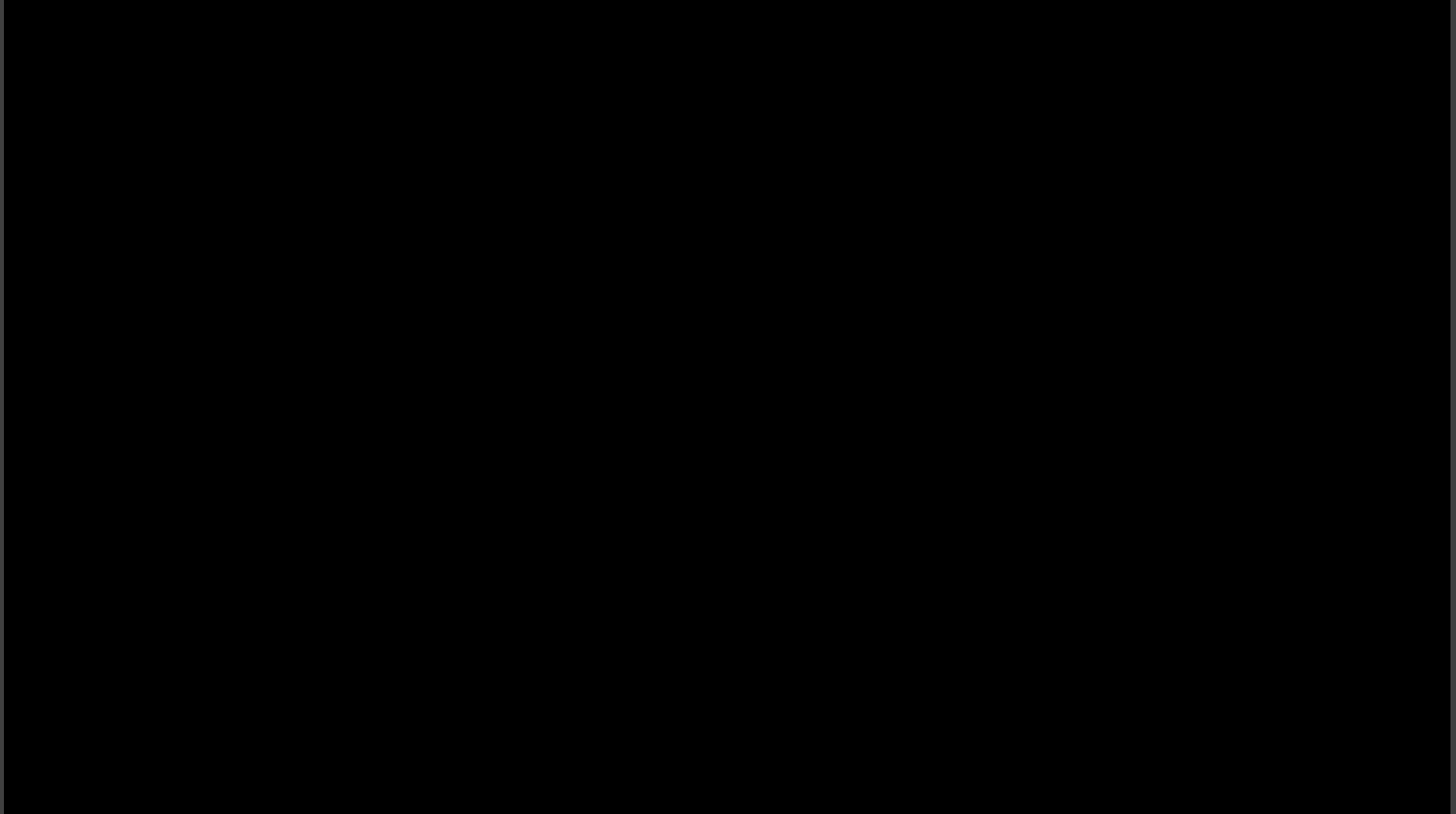
Extrusion



1. Raw material (thermoplastic pellets, granules, powder) are placed into a hopper and fed to a barrel of a *screw extruder*
2. Screw extruder blends and melts raw material (by viscous shearing and external heating) and conveys it towards the die.
3. Pressure builds up
4. Molten plastic is forced through a die

EXTRUSION

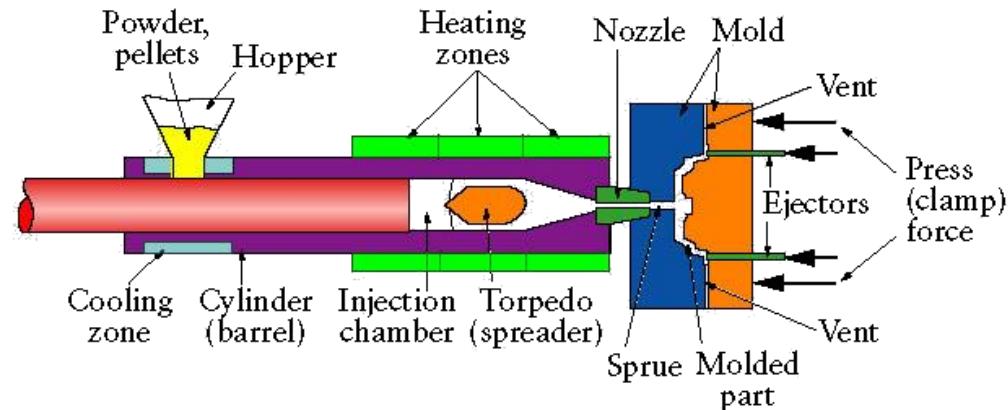
https://www.youtube.com/watch?v=qn16JtE_vLc&ab_channel=ConairGroup



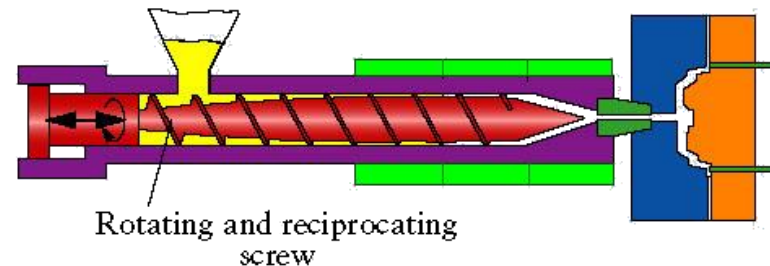
POLYMER PROCESSING

Injection molding

(a)



(b)

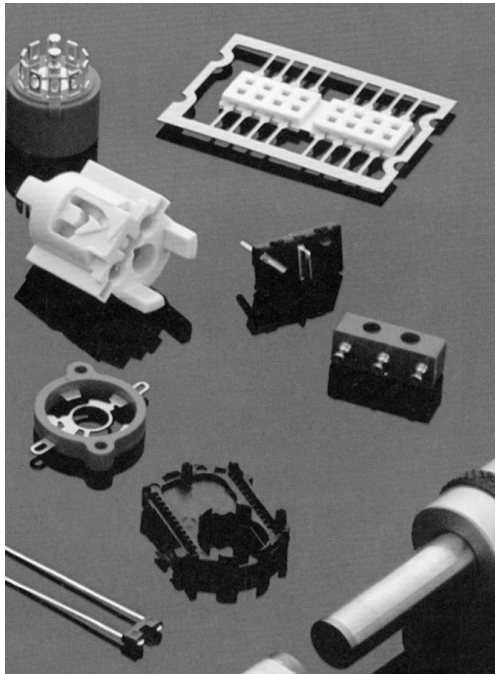


Injection molding with (a) a plunger and (b) a rotating screw.

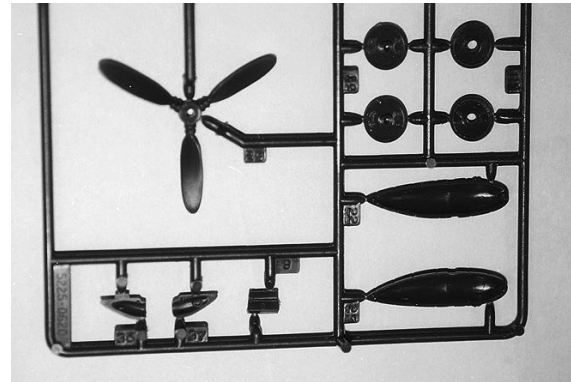
1. Pellets or granules are fed into a heated cylinder
2. Melt is forced into a die by (a) plunger or (b) rotating screw
3. Part is removed from die (after solidification/curing)

POLYMER PROCESSING

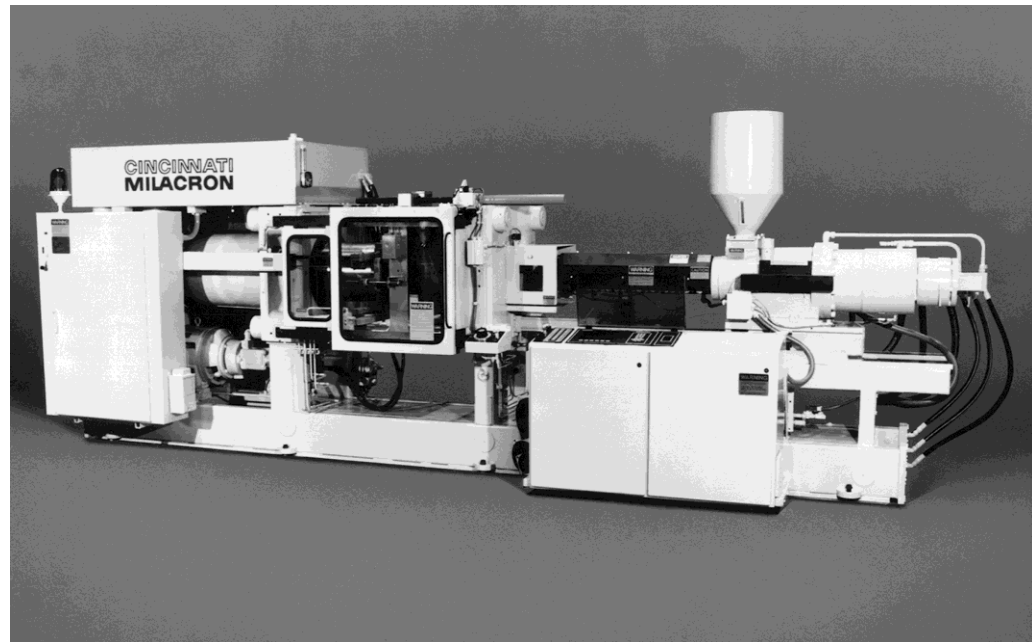
Examples of Injection Molding



Typical products made by injection molding.



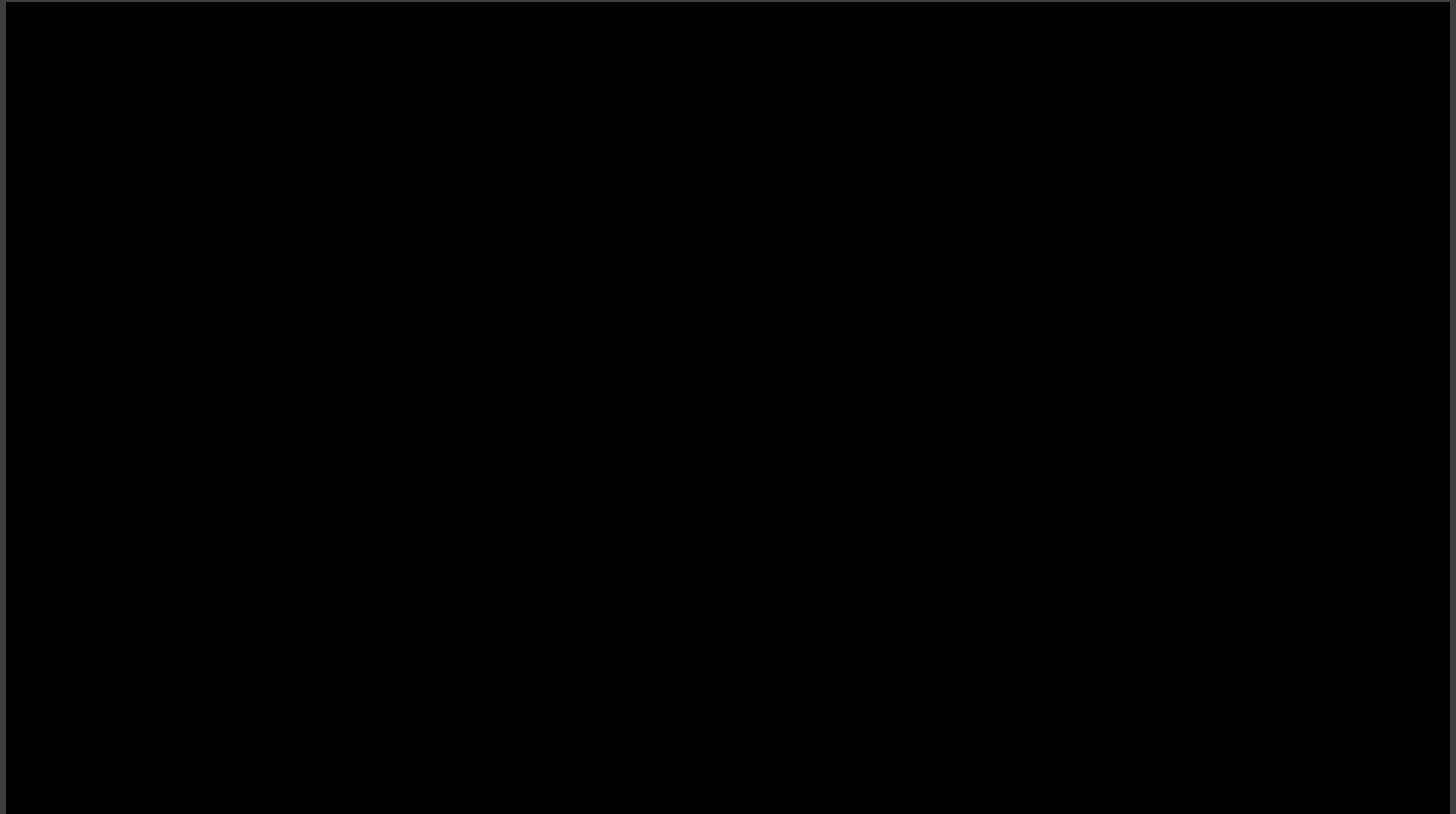
A typical part made from an injection molding machine cavity, showing a number of parts made from one shot; note also mold features such as sprues, runners, and gates.



A 2.2-MN (250-ton) injection-molding machine.

INJECTION

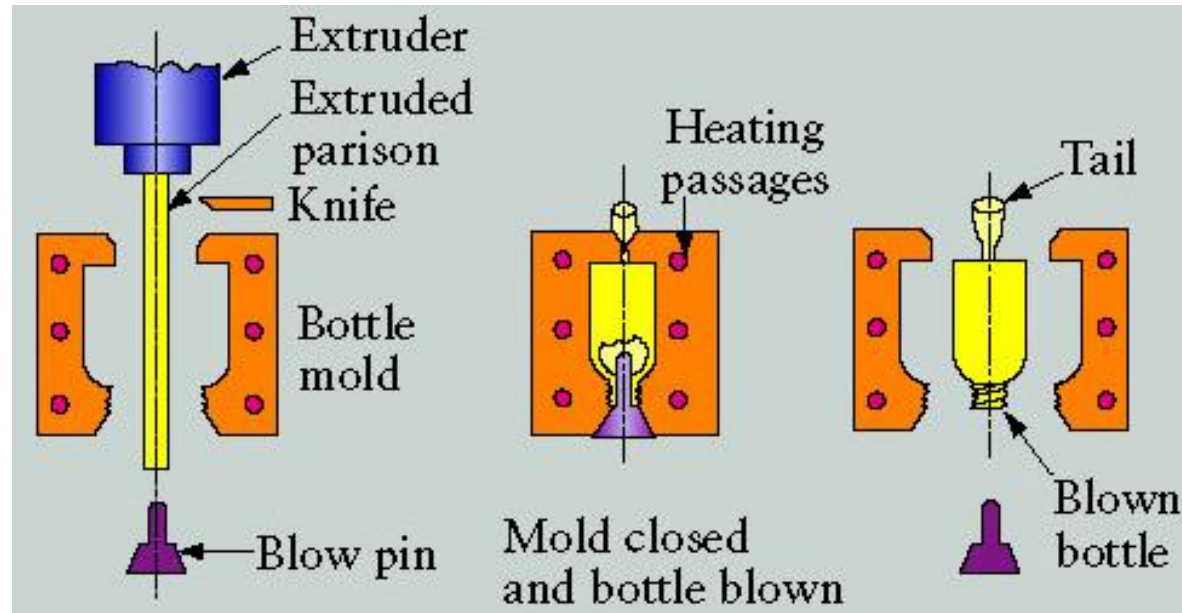
https://www.youtube.com/watch?v=qn16JtE_vLc&ab_channel=ConairGroup



POLYMER PROCESSING

Blow Molding

Schematic illustrations of the *extrusion blow-molding* process for making plastic beverage bottles

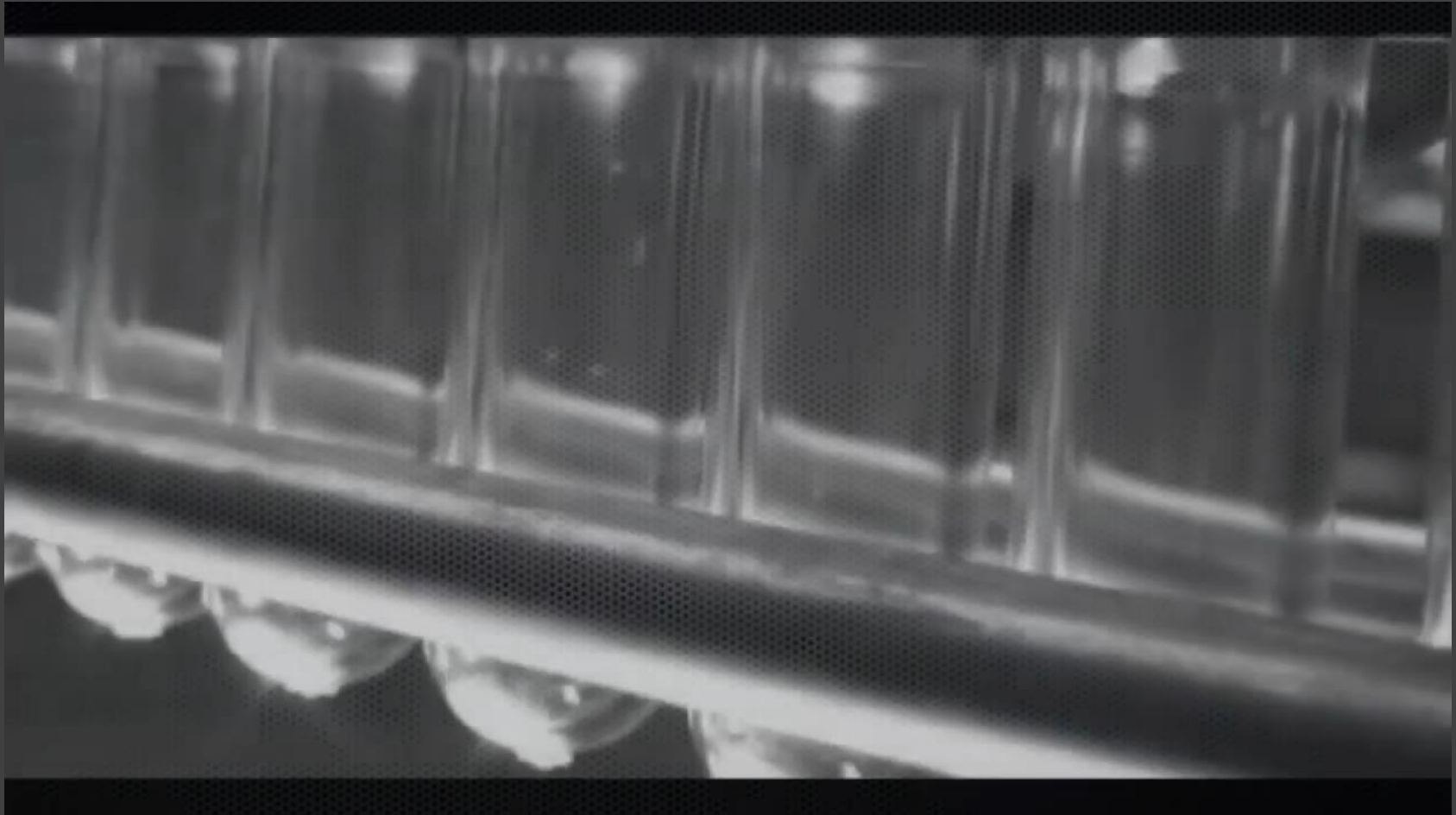


Can be done by *extrusion* or *injection blow molding*:

- Extrusion blow-molding: extruded part is clamped into a larger mold and blown outward to fill the mold cavity
- Injection blow-molding: injection molded part (*parison*) is transferred into blow molding die and hot air is injected and parison is expanded to mold cavity

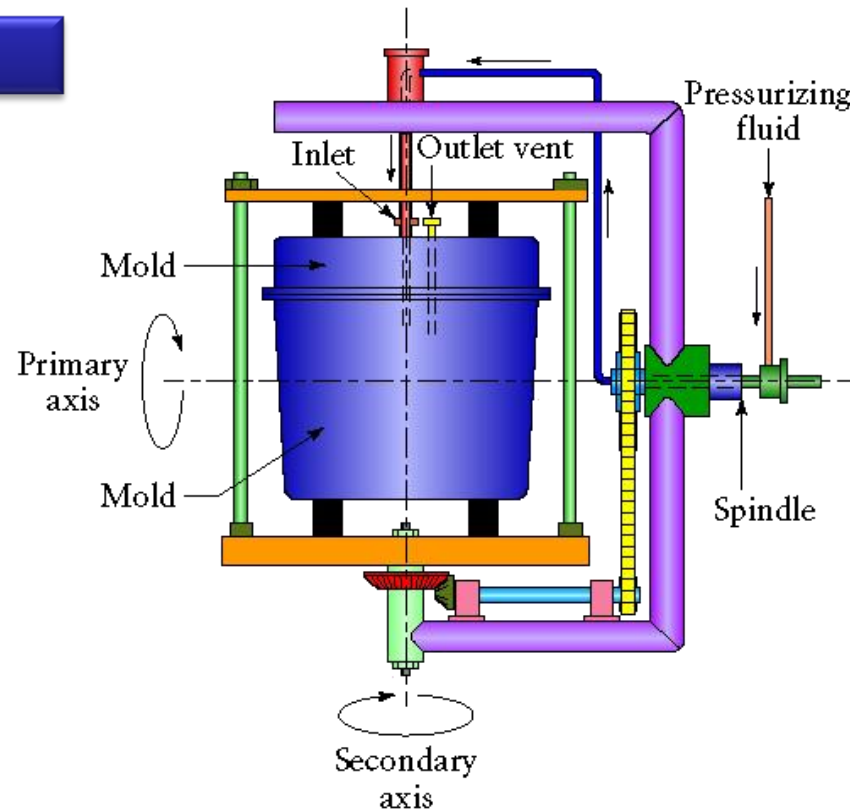
BLOW MOLDING

<https://www.youtube.com/watch?v=NE4c1gwzPb4>



POLYMER PROCESSING

Rotational Molding



1. Powder plastic material is placed into mold
2. Heated mold rotates about its two principal axes
3. Powder fuses against the mold walls

Trash cans, buckets, and plastic footballs can be made by this process

ROTATIONAL MOLDING

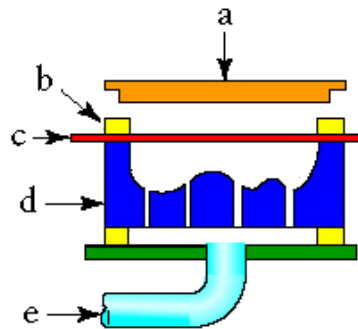
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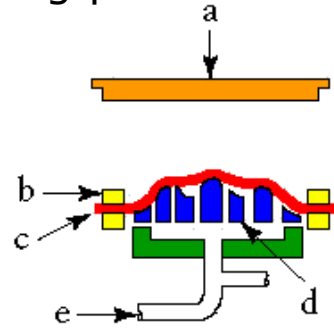
POLYMER PROCESSING

Thermoforming

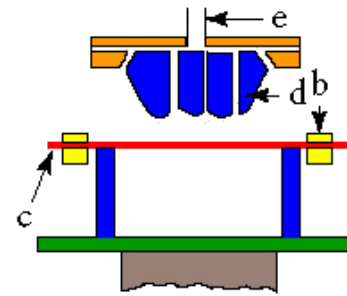
Various thermoforming processes:



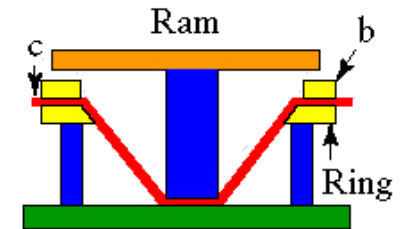
1. Straight vacuum forming



2. Drape vacuum forming



3. Force above sheet



4. Plug and ring forming

a. Heater
b. Clamp
c. Plastic sheet

d. Mold
e. Vacuum line

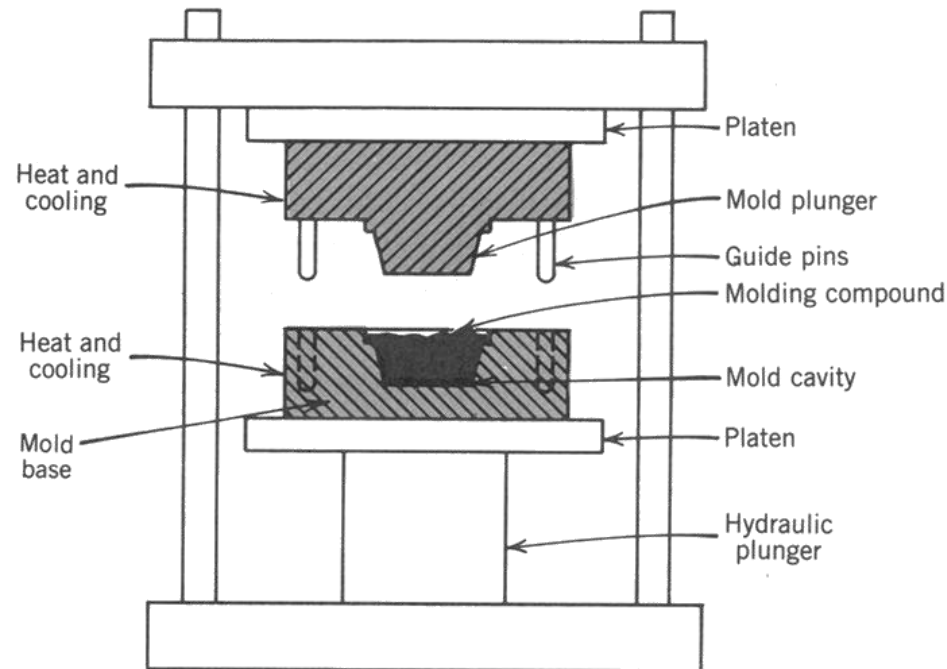
Process for forming for thermoplastic polymers

1. Sheet is clamped and heated above T_g
2. Forced against the mold surfaced by application of vacuum or air pressure

These processes are commonly used in making advertising signs, cookie and candy trays, panels for shower stalls, and packaging

POLYMER PROCESSING

Compression Molding



Similar to die forging. Usually applied to *thermosets*

1. Molding compound (charge of material or pre-measured volume of powder or viscous mixture of thermosetting resin and filler) is placed in heated mold cavity (usually $\sim 200\text{ }^{\circ}\text{C}$)
2. Forming is done by applying Pressure from upper or lower plunger

THERMOFORMING

<https://www.youtube.com/watch?v=SAUpN6zpSpQ>



THERMOFORMING

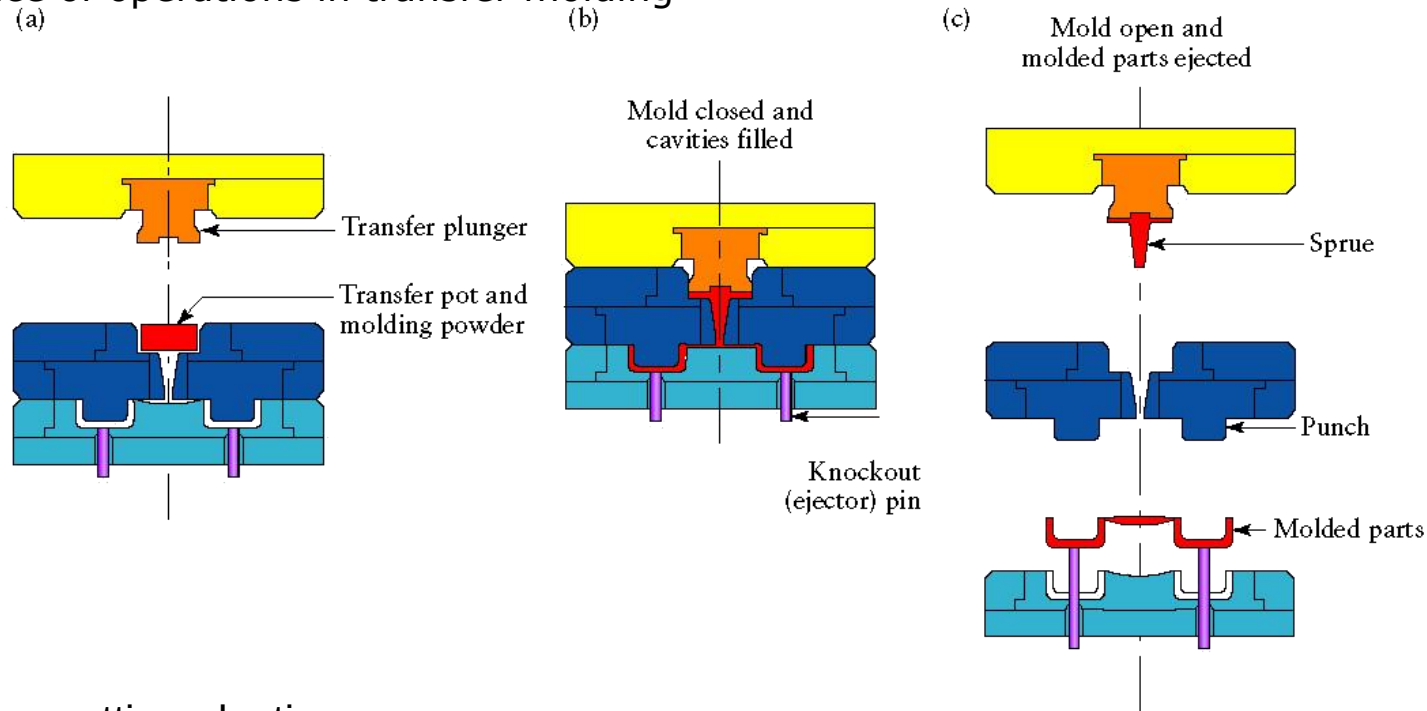
<https://www.youtube.com/watch?v=VyIBsmoiqqs>



POLYMER PROCESSING

Transfer molding

Sequence of operations in transfer molding



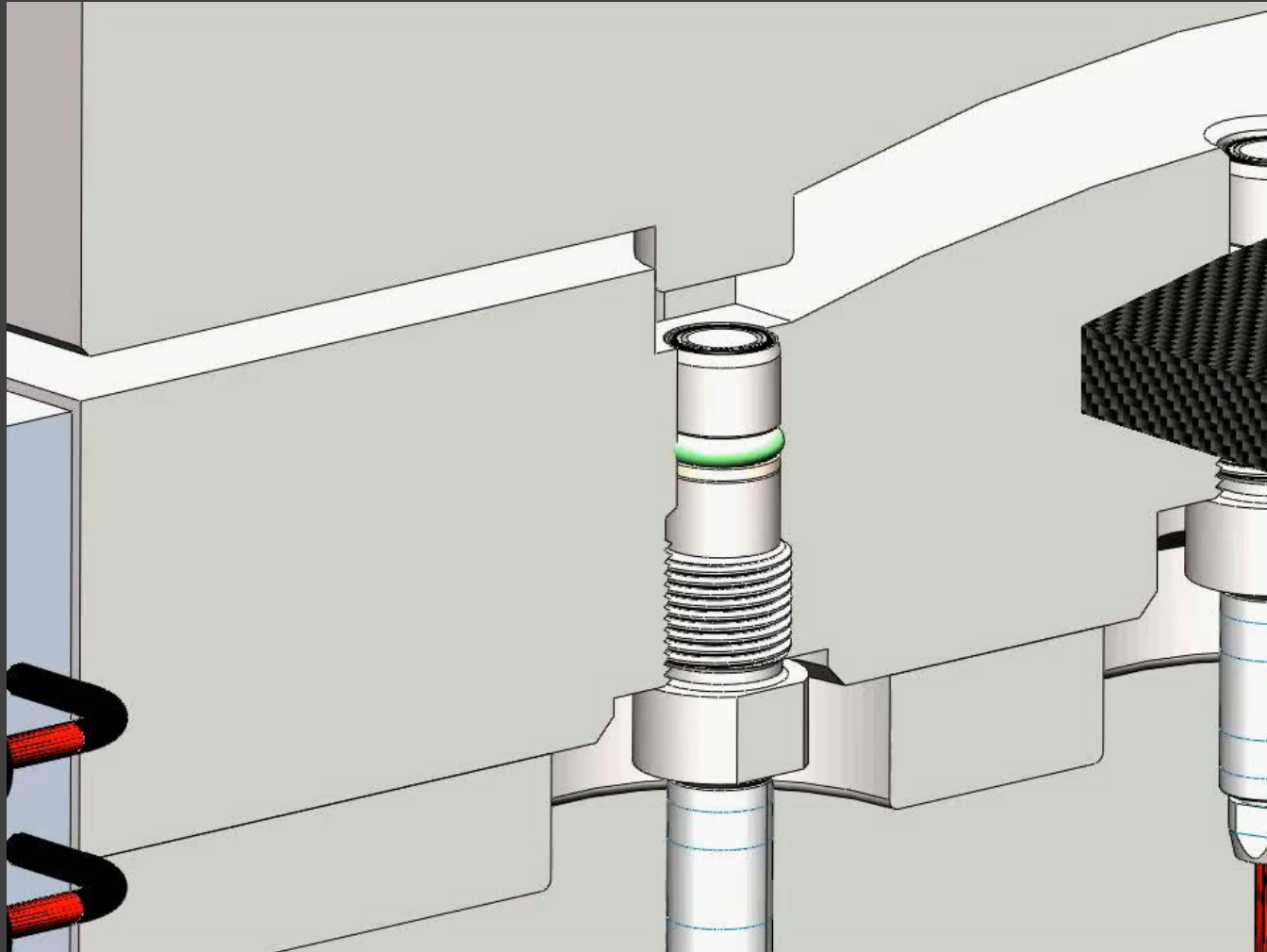
For thermosetting plastics

- Thermosetting resin is placed in heated transfer pot
- Transfer plunger forces the molten resin into mold cavity with pressure
- Setting and cross-linking occurs and molded part is ejected

This process is particularly suitable for intricate parts with varying wall thicknesses.

RESIN TRANSFER MOLDING

https://www.youtube.com/watch?v=NzpHCjL_AnE



POLYMER PROCESSING

Comparative Costs and Production Volumes for Processing of Plastics

	Equipment capital cost	Production rate	Tooling cost	Typical production volume, number of parts						
				10	10 ²	10 ³	10 ⁴	10 ⁵	10 ⁶	10 ⁷
Machining	Medium	Medium	Low							
Compression molding	High	Medium	High							
Transfer molding	High	Medium	High							
Injection molding	High	High	High							
Extrusion	Medium	High	Low	*						
Rotational molding	Low	Low	Low							
Blow molding	Medium	Medium	Medium							
Thermoforming	Low	Low	Low							
Casting	Low	Very low	Low							
Forging	High	Low	Medium							
Foam molding	High	Medium	Medium							

Source: After R.L.E. Brown, *Design and Manufacture of Plastic Parts*. Copyright (c) 1980 by John Wiley & Sons, Inc. Reprinted by permission of John Wiley & Sons, Inc.

*Continuous process.

POLYMER PROCESSING

Characteristics of Forming and Shaping Processes for Plastics and Composite Materials

TABLE 18.1

Process	Characteristics
Extrusion	Long, uniform, solid or hollow complex cross-sections; high production rates; low tooling costs; wide tolerances.
Injection molding	Complex shapes of various sizes, eliminating assembly; high production rates; costly tooling; good dimensional accuracy.
Structural foam molding	Large parts with high stiffness-to-weight ratio; less expensive tooling than in injection molding; low production rates.
Blow molding	Hollow thin-walled parts of various sizes; high production rates and low cost for making containers.
Rotational molding	Large hollow shapes of relatively simple shape; low tooling cost; low production rates.
Thermoforming	Shallow or relatively deep cavities; low tooling costs; medium production rates.
Compression molding	Parts similar to impression-die forging; relatively inexpensive tooling; medium production rates.
Transfer molding	More complex parts than compression molding and higher production rates; some scrap loss; medium tooling cost.
Casting	Simple or intricate shapes made with flexible molds; low production rates.
Processing of composite materials	Long cycle times; tolerances and tooling cost depend on process.

General Recommendations for Plastic Products

Design requirement	Applications	Plastics
Mechanical strength	Gears, cams, rollers, valves, fan blades, impellers, pistons	Acetal, nylon, phenolic, polycarbonate
Functional and decorative	Handles, knobs, camera and battery cases, trim moldings, pipe fittings	ABS, acrylic, cellulosic, phenolic, polyethylene, polypropylene, polystyrene, polyvinyl chloride
Housings and hollow shapes	Power tools, pumps, housings, sport helmets, telephone cases	ABS, cellulosic, phenolic, polycarbonate, polyethylene, polypropylene, polystyrene
Functional and transparent	Lenses, goggles, safety glazing, signs, food-processing equipment, laboratory hardware	Acrylic, polycarbonate, polystyrene, polysulfone
Wear resistance	Gears, wear strips and liners, bearings, bushings, roller-skate wheels	Acetal, nylon, phenolic, polyimide, polyurethane, ultrahigh molecular weight polyethylene

Types de polymers

	Thermoplastics	Thermosets	Elastomers
Heat	Melts	Degrades	Degrades
Solvents	Soluble	Insoluble	Insoluble, they swell
Structure	Linear Branched	Crosslinked	Slightly crosslinked
Cristallinity	Amorphous or crystallinity	Amorphous	Amorphous
Mechanical properties	Rigids at $T < T_g$ $E \approx 10^3$ MPa	Rigids, $\epsilon \approx 4\%$ $E \approx 10^4$ MPa	$\epsilon \approx 100$-1000% E low $\approx$ MPa
Processing	Without chemical reaction	With chemical reaction	With chemical reaction
Examples	PE, PP, PVC, Polyamides, Polyesters	Epoxy resins, Phenol- formaldehyde resins...	Rubber, Polybutadiene, Polyisoprene.

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Questions

1. What is the key difference in the processing of thermoplastic and thermosetting polymers, and how does this difference affect their recyclability?
2. For which type of polymer is compression molding most suitable, and why?
3. Which processing method—extrusion or blow molding—is more appropriate for manufacturing hollow objects like bottles, and why?
4. Why is injection molding often preferred for producing complex, high-volume thermoplastic parts, and why is it not suitable for thermosetting polymers?
5. A product requires precise dimensions, smooth finishes, and high production speed. Which processing method would you choose, and for which type of polymer (thermoplastic or thermosetting)?
6. What type of polymer (thermoplastic or thermoset) is typically used in thermoforming, and why?
7. What processing method would you select for producing continuous lengths of plastic pipes, and why?